

Table 2 Surgical indications for grade III (complete) knee ligament injuries

Ligament injury	Tear type and recommended treatment
ACL	Reconstruction for grade III tears. Repair for bony avulsions from the tibia in pediatric or young adult patients
PCL	Reconstruction for grade III tears with combined injury (i.e., bicruciate, collateral) or repair of bony tibial avulsion fractures
PLC	Reconstruction of grade III injuries and repair of displaced bony avulsions and retracted soft tissue avulsions of the biceps femoris.
MCL	Generally, nonsurgical treatment is performed for isolated grade III injuries. If combined with an ACL tear, may attempt initial nonsurgical treatment. Tibial-based MCL tears are less likely to heal without residual laxity when treated nonsurgically, and reconstruction is recommended.
Multi-ligament	Surgical treatment with reconstruction of cruciate ligaments and reconstruction of collateral/corner injuries. Early intervention is associated with improved outcomes. Staged treatment is associated with higher failure rates.

Abbreviations: ACL, anterior cruciate ligament; MCL, medial collateral ligament; PCL, posterior cruciate ligament; PLC, posterolateral corner.

Multiple graft types and fixation devices are used. Ipsilateral knee autografts, including bone-patellar tendon-bone and hamstring tendons, are commonly used for reconstruction grafts in young patients. Allografts are usually reserved for older patients and those with multiple ligament injuries.

Fixation devices can be grouped based on the location and device type. Interference screws may be used with bone plugs or soft tissue grafts and may be metallic or a radiolucent composite material (►Fig. 9). Alternatively, the graft may be docked in a tunnel with extracortical fixation (i.e., a suture button) (►Fig. 10). Another fixation option includes surgical staples or screw-washer devices adjacent to the nonarticular tunnel aperture (►Fig. 9). Suture anchors may also be utilized for a soft tissue repair or ligament reconstruction (►Fig. 9). Awareness of these fixation options is important for evaluation of postoperative imaging characteristics and potential complications such as loss of fixation with interference screws or inadequate deployment of extracortical buttons.

Senior Author's Preferred Surgical Techniques

The intent of this section is not to serve as a technical manual for surgical treatment but rather to describe surgical indications (►Table 2) as well as key aspects of commonly used techniques and potential postoperative imaging findings

(►Table 3). Full surgical technique details are available in the specifically referenced articles (e.g., tunnel size, knee fixation angle, order of graft fixation). Importantly, the native anatomy guides the desired tunnel locations, and a thorough understanding of the native anatomy is important for a review of postoperative imaging.

Anterior Cruciate Ligament Reconstruction

Reconstruction of the ACL is performed with a patellar tendon autograft. Anatomical femoral and tibial tunnels¹⁰ are created, and the graft is generally secured with titanium interference screws in the femur and tibia (►Fig. 9). The femoral tunnel should be positioned distal to the Blumensaat line on lateral radiographs,²⁸ and the tibial tunnel should be adjacent to the anterior root attachment of the lateral meniscus on axial MRI sections (►Fig. 4). Staples may be utilized to secure the bone plug in a trough in cases of graft-tunnel mismatch; occasionally one prong of the staple may skive along the tibial shaft, which has no implications for the clinical outcome.

Posterior Cruciate Ligament Reconstruction

An arthroscopic-assisted double-bundle PCL reconstruction recreates the anterolateral bundle (ALB) with an Achilles tendon allograft and the posteromedial bundle (PMB) with a

Table 3 Postoperative radiographic and magnetic resonance imaging pearls

Finding	Possible clinical scenario
Femoral and/or tibial tunnel widening on radiographs or MRI	Compromised integrity with graft tears that may require a two-stage revision with tunnel bone grafting followed by ligament reconstruction
Anteriorly placed PCL reconstruction tibial tunnel	Iatrogenic injury to the meniscal root(s)
Arthrofibrosis	Limited motion and terminal extension
Graft impingement	Ligamentous laxity or instability and early failure; cyclops lesion
Subluxation on radiograph or MRI	May indicate graft laxity; stress radiographs should be obtained to quantify laxity

Abbreviations: MRI, magnetic resonance imaging; PCL, posterior cruciate ligament.